Inflectional vs Derivational Morphology in NLP

Characteristics, applications and challenges

walk → walked
Inflectional: new form, same word

happy → unhappiness
Derivational: new word, new meaning

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What is morphology?



prefix | root / stem | suffix

un + break + able = unbreakable

Free morpheme

Can stand alone as a word: break, happy, teach.

Bound morpheme

Must attach to another: un-, -able, -ed, -ness.

Inflectional morphology

- Adds grammar: tense, number, case, degree
- Same word class and core meaning
- Regular and highly productive
- Always the outermost layer of a word

Suffix	Role	Example
-s	plural	cat → cats
-s	3rd person	walk → walks
-ed	past tense	walk → walked
-ing	progressive	walk → walking
-er / -est	comparison	big → bigger

Derivational morphology

- Creates a new word (new lexeme)
- Often changes the word class
- Meaning can shift or drift
- Selective: many gaps and exceptions

verb → noun
teach → teacher

adj → noun
happy → happiness

noun → adj
care → careful

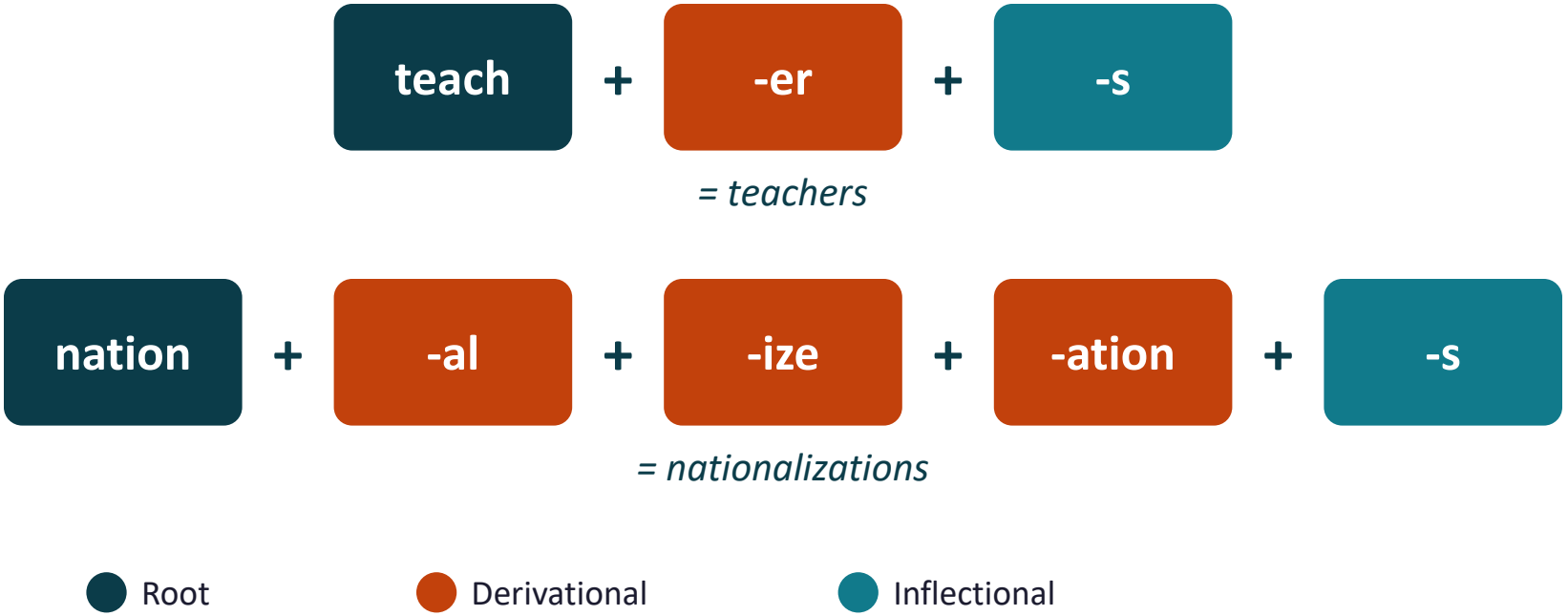
adj → verb
active → activate

Side-by-side comparison

Aspect	Inflectional	Derivational
Purpose	Marks tense, number, case	Builds new words
Word class	Unchanged: walk → walked	Often changes: teach → teacher
Meaning	Core meaning unchanged	New or shifted meaning
Productivity	Regular, applies broadly	Selective, many exceptions
Affix order	Outer layer (teach-er-s)	Inner layer (teach-er)
Examples	-s, -ed, -ing, -est	-ness, -able, un-, re-

Rule of thumb: inflection changes the form, derivation changes the word.

How they combine



Applications in NLP

I Inflectional

- Lemmatization and stemming for search and indexing
- Agreement checks in POS tagging and parsing
- Correct word forms in machine translation
- Text normalization before tokenization

D Derivational

- Vocabulary and query expansion
- Sentiment cues from affixes (un-, -less)
- Guessing the meaning of unseen words
- Subword models such as BPE and WordPiece

Challenges

Inflectional

Irregular forms

went, mice and better break rule-based stemmers

Derivational

Ambiguous affixes

-er in teacher vs bigger; un- in undo vs unhappy

Derivational

Unpredictable meaning

department is not depart + ment; meaning drifts

Both

Data sparsity

Rich morphology (Tamil, Turkish) explodes the vocabulary

Techniques used in NLP

1. **Rule-based stemmers**
   Porter, Snowball: chop suffixes by rules. Fast, but crude.

2. **Dictionary lemmatizers**
   WordNet, spaCy: return the real base form using POS.

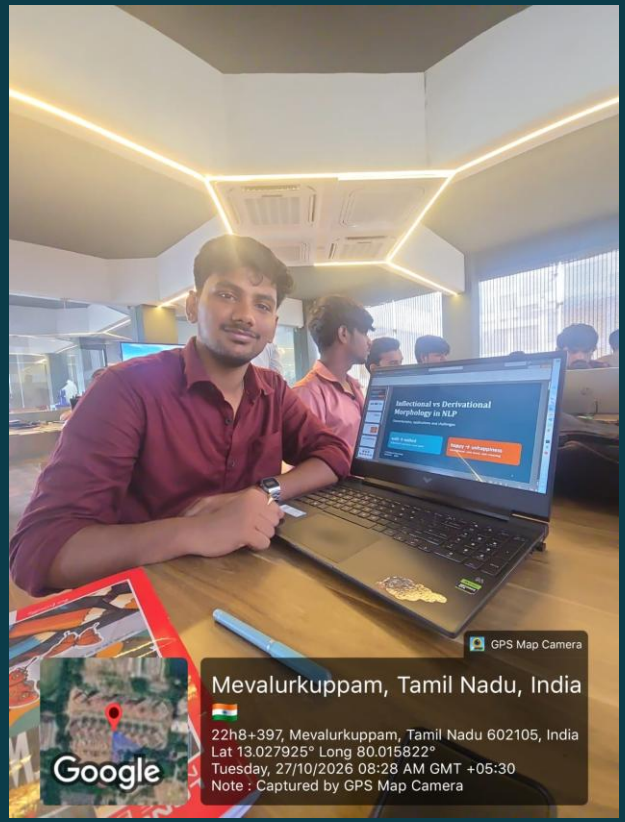
3. **Finite-state analyzers**
   Full analysis of rich morphology, e.g. Turkish, Tamil.

4. **Subword tokenizers**
   BPE, WordPiece: learn word pieces from data.

Key takeaways

1. Inflection changes the form; derivation creates a new word.
2. Inflection is regular and rule-friendly; derivation needs semantics.
3. Modern NLP combines morphological analyzers with subword models to handle both.

Presentation



THANK YOU!!